CS202 Fundamentals of Front-end Development

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1. HTML is local storage? What benefits HTML5 provides over HTML?
2. Word-wrap property? Syntax code?
3. If table was given. How to write fadeIn() and fadeOut()?
4. Write a code for an image with width 100 and height auto.
5. Write advantages of XML namespace.
6. What are semantic elements? Give examples.
7. Write JavaScript global functions with properties.
8. Write the three audio formats that HTML5 supports.
9. Write the syntax to set multiple images in the background in CSS3.
10. Why do we need XML DDT? Explain briefly.
11. Write the code for responsive website, but the page of 2 columns.
12. Define HTML5 local storage and what benefits does HTML5 provides over other previous HTML versions?
13. XML Attributes with Example?
14. Note on jQuery Selectors.
15. Write Parts of A URL With Two Example.
16. Write the Code for Multiple images for different Background Sizes.
17. Define Word-wrap Property with Example Syntax Code.
18. Briefly explain 'lists' in HTML and types of lists in.
19. FadeIn, FadeOut type, XML, xpath, DTD, DOM stands for? <bdi> stands for?
20. Define XML attributes problems.
21. Write a code for image with width 100 and height auto.
22. Write CSS in an HTML code.
23. Define $("p"), $("ul: li first"), $(ul: li first-child)
24. Given a picture with one red and one blue block. Write the CSS code.
25. Define $("*"), $(this), $("p.test"), $(“p:first”)
26. Define <mark>, <figcaption>, <nav>, <footer>, <bdi> in HTML.
27. Write a code for 2 columns in web responsive.
28. Types of gradients in CSS3?
29. Enlist jQuery methods to manipulate CSS.
30. Problems of using attributes in XML?
31. Difference b/w SVG and Canvas?
32. Five methods of jQuery to traverse siblings in DOM tree?
33. Write code to generate RWD which have two columns.
34. Three useful methods of jQuery for traversing up DOM tree.
35. What is meant by RWD (Responsive Web Design) or Grid view?
36. What is meant by <!DOCTYPE html>.
37. Enlist three same characters use in HTML4 and HTML5.
38. HTML5 and HTML4 necessary for proper web page?
39. jQuery three methods for CSS manipulation.
40. Write three different attributes of JSON from XML.
41. If table is given and you have to write fadeIn.
42. Two parameters was given speed, ease and you have to write its default and possible value.
43. What is Wrapper Class?
44. Difference between break and continue in JavaScript?
45. Difference between array and object?
46. Write Shadow-text code.
47. Enlist XML attributes.
48. Write the syntax of animate in jQuery.
49. Semantic elements and it's three new elements?
50. Events in XML?

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Answers:

1. HTML Local Storage and Benefits of HTML5 Over HTML

- **Local Storage**: It is part of the Web Storage API in HTML5, used to store key-value pairs locally within the browser, with no expiration time. Data remains even if the page is closed and reopened.

Example:

```
localStorage.setItem('username', 'John');
console.log(localStorage.getItem('username')); // Output: John
```

- **Benefits of HTML5 Over HTML**:
    1. **Native Multimedia**: Built-in support for audio and video tags without external plugins.
    2. **Semantic Tags**: New elements like <article>, <section>, <header> improve readability and SEO.
    3. **APIs**: Added Geolocation, WebSockets, and Web Storage for modern app needs.

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2. Word-Wrap Property

- **Definition**: It forces text to break onto the next line when it is too long to fit within its container.
- **Syntax**:

```
word-wrap: break-word;
```

- **Example**:

```
<div style="width: 100px; border: 1px solid black; word-wrap: break-word;">
  ThisIsAVeryLongWordThatWillBreak
</div>
```

---

3. FadeIn() and FadeOut() in jQuery

- **fadeIn()**: Gradually increases the opacity of an element from 0 to 1.
- **fadeOut()**: Gradually decreases the opacity of an element from 1 to 0.

Example:

```
$("table").fadeIn(500); // Fades in over 500ms
$("table").fadeOut(500); // Fades out over 500ms
```

---

4. Code for an Image with Width 100 and Height Auto

- **Explanation**: The width is fixed at 100px, while the height adjusts proportionally to maintain the aspect ratio.
- **Code**:

```
<img src="image.jpg" style="width: 100px; height: auto;">
```

---

5. Advantages of XML Namespace

1. **Avoids Conflicts**: Prevents name clashes in documents using multiple vocabularies.
2. **Extensibility**: Allows combining different XML vocabularies in a single document.
3. **Uniqueness**: Provides a unique identity to elements using URIs.

Example:

```
<svg xmlns="http://www.w3.org/2000/svg">
  <circle cx="50" cy="50" r="40" />
</svg>
```

---

6. Semantic Elements in HTML

- **Definition**: Elements that provide meaning to content, helping search engines and developers understand the structure of a webpage.
- **Examples**:
    - <header>: Represents the page header.
    - <footer>: Defines the footer section.
    - <nav>: Represents navigation links.

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7. JavaScript Global Functions

1. **parseInt()**: Converts strings to integers.

```
parseInt("42"); // Output: 42
```

2. **isNaN()**: Checks if a value is not a number.

```
isNaN("abc"); // Output: true
```

3. **eval()**: Executes a string as JavaScript code.

```
eval("2 + 2"); // Output: 4
```

---

8. Audio Formats Supported by HTML5

1. **MP3 (audio/mpeg)**: Universal compatibility.
2. **Ogg (audio/ogg)**: Open-source and efficient.

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3. **WAV (audio/wav)**: High-quality but larger file size.

Example:

```
<audio controls>
  <source src="audio.mp3" type="audio/mpeg">
  <source src="audio.ogg" type="audio/ogg">
  Your browser does not support the audio tag.
</audio>
```

---

9. Syntax for Multiple Background Images in CSS3

- **Explanation**: Multiple background images can be layered, with each image having its own position and size.
- **Code**:

```
body {
  background-image: url('image1.jpg'), url('image2.jpg');
  background-position: top left, bottom right;
  background-size: cover, contain;
}
```

---

10. Why Do We Need XML DTD?

- **Definition**: Document Type Definition (DTD) defines the structure, elements, and attributes of an XML document.
- **Importance**:
  1. Ensures XML documents follow a specific format.
  2. Helps validate XML against predefined rules. Example:

```
<!DOCTYPE note [
  <!ELEMENT note (to, from, heading, body)>
  <!ELEMENT to (#PCDATA)>
]>
```

---

11. Code for a Responsive Website with Two Columns

- **Explanation**: Use CSS Flexbox to create a responsive layout.
- **Code**:

```
<div style="display: flex; flex-wrap: wrap;">
  <div style="flex: 1; min-width: 200px; background: lightblue;">Column 1</div>
  <div style="flex: 1; min-width: 200px; background: lightgreen;">Column 2</div>
</div>
```

---

12. Benefits of HTML5 Local Storage

- **Definition**: Stores data locally in the browser in key-value pairs.
- **Benefits**:
  1. **Persistent Storage**: Data remains after closing the browser.
  2. **Increased Storage Capacity**: Up to 5MB compared to cookies.
  3. **Reduces Server Load**: No server-side involvement.

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13. XML Attributes with Example

- **Definition**: Provide additional information about elements in key-value pairs.
- **Example**:

```
<book title="XML Basics" author="John Doe" />
```

14. Note on jQuery Selectors

- **Definition**: jQuery selectors allow you to select and manipulate HTML elements based on their tags, classes, IDs, or attributes.
- **Common Selectors**:
  1. **$("*")**: Selects all elements.
  2. **$("p")**: Selects all <p> elements.
  3. **$(".class")**: Selects elements with the specified class.
  4. **$("#id")**: Selects the element with the specified ID.

Example:

```
$("p").css("color", "blue"); // Changes text color of all <p> elements to blue.
```

---

15. Parts of a URL

- **Parts of a URL**:

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1. **Scheme**: Protocol used (e.g., https://).
2. **Host**: Domain name (e.g., www.example.com).
3. **Path**: Specific location/resource on the server (e.g., /index.html).
4. **Query String**: Parameters sent to the server (e.g., ?id=123).
5. **Fragment**: Internal page reference (e.g., #section1).

Examples:

- Full URL: https://www.example.com/products?id=123#details
- Components:
  - Scheme: https
  - Host: www.example.com
  - Path: /products
  - Query: ?id=123
  - Fragment: #details

---

16. Code for Multiple Images for Different Background Sizes

- **Explanation**: You can apply different background sizes and positions to each image.
- **Code**:

```
body {
  background-image: url('small.jpg'), url('large.jpg');
  background-size: 50px 50px, cover;
  background-position: top left, center;
}
```

---

17. Word-Wrap Property with Example

- **Definition**: Forces text to break onto the next line if it's too long to fit within the container.
- **Syntax**:

```
word-wrap: break-word;
```

- **Example**:

```
<div style="width: 150px; border: 1px solid black; word-wrap: break-word;">
  ThisIsAVeryLongWordThatWillBreak
</div>
```

---

18. Lists in HTML and Their Types

- **Definition**: Lists are used to group related items together.
- **Types**:
  1. **Ordered List (<ol>)**: Numbered items.
  2. **Unordered List (<ul>)**: Bulleted items.
  3. **Description List (<dl>)**: Key-value pairs.

Example:

```
<ol>
  <li>Step 1</li>
  <li>Step 2</li>
</ol>
<ul>
  <li>Apple</li>
  <li>Banana</li>
</ul>
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language</dd>
</dl>
```

---

19. Abbreviations and Their Meanings

- **DOM**: Document Object Model (structure of a webpage).
- **DTD**: Document Type Definition (defines XML structure).
- **bdi**: Bi-Directional Isolation (isolates text direction in different languages).
- **XML**: Extensible Markup Language (data transport format).
- **XPath**: XML Path Language (query XML data).

---

20. XML Attributes Problems

- **Definition**: Attributes provide metadata about an element but are limited.
- **Problems**:
  1. Cannot store hierarchical data.

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2. Harder to parse compared to elements.
3. Cannot include multiple values.

Example Problem:

<person name="John" age="30" />

- Better Alternative:

<person>
 <name>John</name>
 <age>30</age>
</person>

---

21. Code for Image with Width 100 and Height Auto

- **Code**:

<img src="image.jpg" style="width: 100px; height: auto;">

- **Explanation**: This keeps the image proportionate while ensuring the width is 100px.

---

22. CSS in HTML

- **Definition**: CSS can be written inside <style> tags in the HTML <head> section.
- **Example**:

```
<!DOCTYPE html>
<html>
<head>
 <style>
  body {
   background-color: lightblue;
  }
  p {
   color: blue;
   font-size: 20px;
  }
 </style>
</head>
<body>
 <p>This is styled text.</p>
</body>
</html>
```

---

23. jQuery Selectors Explanation

1. **$("p")**: Selects all <p> elements.
2. **$("ul li:first")**: Selects the first <li> inside each <ul>.
3. **$("ul li:first-child")**: Selects <li> elements that are the first child of their parent.

Example:

$("ul li:first").css("color", "red"); // Changes the first <li> in each <ul> to red.

---

24. CSS Code for Red and Blue Blocks

- **Code**:

```
<div class="red"></div>
<div class="blue"></div>

<style>
 .red {
  background-color: red;
  width: 100px;
  height: 100px;
  float: left;
 }
 .blue {
  background-color: blue;
  width: 100px;
  height: 100px;
  float: left;
  margin-left: 10px;
 }
</style>
```

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25. jQuery Selectors

- **$("p.test")**: Selects all <p> elements with class="test".
- **$("*")**: Selects all elements in the DOM.
- **$(this)**: Refers to the current HTML element in a specific event.
- **$("p:first")**: Selects the first <p> element.

26. Semantic Elements in HTML

- **Definition**: Semantic elements clearly define their purpose and the type of content they hold, improving readability and accessibility.
- **Examples**:
    - <mark>: Highlights text for emphasis.

```
<p>The word <mark>important</mark> is highlighted.</p>
```

    - <figcaption>: Provides a caption for a figure.

```
<figure>
 <img src="image.jpg" alt="Example">
 <figcaption>An example image.</figcaption>
</figure>
```

    - <nav>: Contains navigation links.

```
<nav>
 <a href="home.html">Home</a>
 <a href="about.html">About</a>
</nav>
```

    - <footer>: Represents the footer of a document or section.
    - <bdi>: Isolates text direction for mixed-language content.

```
<p>The user <bdi>محمد</bdi> logged in.</p>
```

---

27. Responsive Design Code for Two Columns

- **Explanation**: A responsive two-column layout adjusts automatically based on screen size using media queries.
- **Code**:

```
<div class="container">
 <div class="column">Column 1</div>
 <div class="column">Column 2</div>
</div>

<style>
 .container {
  display: flex;
  flex-wrap: wrap;
 }
 .column {
  flex: 1;
  min-width: 300px;
  padding: 10px;
  border: 1px solid black;
 }
</style>
```

---

28. Types of Gradients in CSS3

1. **Linear Gradient**: Colors transition along a straight line.

```
background: linear-gradient(to right, red, blue);
```

2. **Radial Gradient**: Colors transition outward from a center point.

```
background: radial-gradient(circle, red, yellow, green);
```

3. **Conic Gradient**: Colors transition around a circle.

```
background: conic-gradient(red, yellow, green);
```

---

29. jQuery Methods to Manipulate CSS

1. **.css()**: Sets or retrieves a CSS property.

```
$("p").css("color", "blue");
```

2. **.addClass()**: Adds a class to selected elements.

```
$("div").addClass("highlight");
```

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3. **.removeClass()**: Removes a class from elements.

```
$("div").removeClass("highlight");
```

---

30. Problems of Using Attributes in XML

1. **Limited Data**: Attributes cannot store complex or hierarchical data.
2. **Reduced Readability**: Elements are easier to read than attributes.
3. **Lack of Scalability**: Attributes do not support multiple values.

---

31. Difference Between SVG and Canvas

Feature	SVG	Canvas
Definition	Scalable Vector Graphics (XML-based).	Bitmap-based 2D graphics.
Resolution	Scales without losing quality.	Pixelated when scaled.
Manipulation	DOM elements, easily styled.	Requires JavaScript for changes.
Example	<circle cx="50" cy="50" r="40" />	context.arc(50, 50, 40, 0, 2 * Math.PI);

---

32. Five Methods to Traverse Siblings in DOM Tree

1. **.next()**: Selects the next sibling.

```
$("li").next();
```

2. **.prev()**: Selects the previous sibling.

```
$("li").prev();
```

3. **.siblings()**: Selects all siblings.

```
$("li").siblings();
```

4. **.nextAll()**: Selects all next siblings.

```
$("li").nextAll();
```

5. **.prevAll()**: Selects all previous siblings.

```
$("li").prevAll();
```

---

33. Code for RWD with Two Columns

- **Code** (same as #27):

```
<div class="container">
 <div class="column">Column 1</div>
 <div class="column">Column 2</div>
</div>

<style>
 .container {
  display: flex;
  flex-wrap: wrap;
 }
 .column {
  flex: 1;
  min-width: 300px;
  padding: 10px;
  border: 1px solid black;
 }
</style>
```

---

34. Useful Methods for Traversing Up DOM Tree

1. **.parent()**: Selects the immediate parent.

```
$("span").parent();
```

2. **.parents()**: Selects all ancestors.

```
$("span").parents();
```

3. **.parentsUntil()**: Selects ancestors up to a specified element.

```
$("span").parentsUntil("div");
```

---

35. Responsive Web Design (RWD) or Grid View

- **RWD**: A design approach ensuring websites look good on all devices by using flexible grids, media queries, and layouts.
- **Grid View**: Divides a page into rows and columns for easier layout management.

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36. <!DOCTYPE html>

- **Definition**: Declares the document type and version of HTML.
- **Purpose**:
    1. Ensures the browser interprets the document as HTML5.
    2. Prevents quirks mode in older browsers.

Example:

```
<!DOCTYPE html>
<html>
<head></head>
<body></body>
</html>
```

---

37. Characters in Both HTML4 and HTML5

1. **&**: Represents &.
2. **<**: Represents <.
3. **>**: Represents >.

---

38. HTML4 and HTML5 for Proper Web Page

- **HTML4**: Lacks modern features like multimedia and APIs.
- **HTML5**: Enhances user experience with semantic elements, multimedia, and offline capabilities.

---

39. jQuery Methods for CSS Manipulation

1. **.css()**: Sets or gets CSS properties.
2. **.addClass()**: Adds CSS classes.
3. **.removeClass()**: Removes CSS classes.

---

40. JSON vs. XML Attributes

1. JSON uses key-value pairs, while XML uses attributes.
2. JSON is lighter and faster for data exchange.
3. JSON syntax is simpler.

---

41. FadeIn with Speed and Ease

- **Code**:

```
$("table").fadeIn({
  duration: 400,   // Speed (ms)
  easing: "swing" // Default easing
});
```

---

42. Wrapper Class

- **Definition**: A class used to wrap other elements, applying common styles or behaviors.

Example:

```
<div class="wrapper">
  <p>Content</p>
</div>
```

---

43. Break vs. Continue in JavaScript

- **Break**: Exits the loop immediately.
- **Continue**: Skips the current iteration and moves to the next.

Example:

```
for (let i = 0; i < 5; i++) {
  if (i === 2) break;   // Stops at 2
  if (i === 1) continue; // Skips 1
}
```

44. Difference Between Array and Object

- **Array**: A collection of ordered elements.
- **Object**: A collection of key-value pairs.

Feature	Array	Object
Structure	Indexed (e.g., arr[0]).	Key-value pairs (e.g., obj.key).
Use Case	Used for lists of data.	Used for structured data.
Syntax	let arr = [1, 2, 3];	let obj = {name: "John"};
Access	Accessed by index.	Accessed by key.

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Example:
// Array
let fruits = ["apple", "banana", "cherry"];
console.log(fruits[0]); // Output: apple

// Object
let person = { name: "Alice", age: 25 };
console.log(person.name); // Output: Alice

---

45. Write Shadow-Text Code

- **Explanation**: The text-shadow property in CSS adds shadow effects to text.
- **Code**:

<h1 style="text-shadow: 2px 2px 5px gray;">Shadow Text</h1>

- **Output**: The text appears with a gray shadow offset by 2px horizontally and vertically, with a blur radius of 5px.

---

46. Enlist XML Attributes

- **Definition**: Attributes in XML store metadata for elements.
- **Examples**:
    1. <book title="XML Guide" author="John" />
    2. <employee id="101" department="HR" />

---

47. Syntax of animate() in jQuery

- **Definition**: The animate() method performs custom animations on CSS properties.
- **Syntax**:

$(selector).animate(properties, duration, easing, complete);

- **Parameters**:
    1. **properties**: CSS styles to animate.
    2. **duration**: Speed of the animation (fast, slow, or milliseconds).
    3. **easing**: Animation effect (swing or linear).
    4. **complete**: Callback function executed after the animation.

Example:

```
$("div").animate({
  width: "300px",
  height: "200px"
}, 1000, "swing", function() {
  alert("Animation Complete");
});
```

---

48. Semantic Elements and Their Three New Elements

- **Semantic Elements**: Clearly describe their meaning in a human-readable way.
- **Examples of New Semantic Elements in HTML5**:
    1. <article>: Represents a standalone piece of content.
    2. <section>: Groups related content.
    3. <aside>: Represents content tangentially related to the main content.

---

49. Events in XML

- **Definition**: XML itself does not have events, but when processed (e.g., in a browser or parser), events like these can occur:
    1. **Start Element**: Triggered when a new element starts.
    2. **End Element**: Triggered when an element closes.
    3. **Character Data**: Triggered when content inside an element is read.

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